CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The automobile industry is one of the biggest contributors to India's manufacturing output and employment. A car or truck maker has to spend heavily on plants, research and new technology, and it has to do so while customer demand, raw material prices and government rules keep changing. Because of this, how well a company manages its money matters as much as how many vehicles it sells. Financial performance shows whether a company earns enough from its business, keeps its costs under control and produces enough cash to grow.

Tata Motors is a good company to study from this point of view. It is one of India's largest vehicle makers, and through Jaguar Land Rover it also sells luxury vehicles in many countries. In the last few years its results have changed a lot. The company went through a difficult period with the pandemic and the global chip shortage, and then reported record revenue and profit in FY2024 (Tata Motors, 2024).

Companies today also produce a huge amount of data every day, in the form of sales records, cost sheets and cash flow statements. Reading such data line by line is slow and easy to get wrong. Business analytics tools help to clean the data, organise it, ask questions of it and show the answers in charts. This study uses three common tools for that purpose: Microsoft Excel, SQL and Tableau. They are applied to a dataset of Tata Motors sales and financial records for the financial years 2022 to 2024.

1.2 Overview of Tata Motors

History, Vision and Mission

Tata Motors began in 1945 as a part of the Tata Group, whose roots go back to Jamsetji Tata. It started with locomotives and then moved into commercial vehicles. Over the following decades the company widened its range. It later entered the car market, and in 1998 it launched the Indica, which was widely described as India's first car designed and built in the country. Brands such as the Sumo, Safari and Indigo followed, and in 2008 the company unveiled the Nano, a very low priced car (Divya & Abirami, 2016; Trivedi, 2014).

The company describes its purpose in terms of mobility that improves people's quality of life, together with a focus on cleaner technology and the values followed across the Tata Group, such as integrity and responsibility.

Major Products and Business Segments

Tata Motors works mainly in two areas. The first is passenger vehicles, which includes cars, SUVs and electric vehicles in India, and the luxury brands Jaguar and Land Rover. The second is commercial vehicles, which covers trucks, buses and small commercial vehicles used for goods and passenger transport.

The structure of the company changed in 2025. The demerger became effective on 1 October 2025, and the two businesses now operate as separate listed companies. The passenger vehicle business, along with electric vehicles and Jaguar Land Rover, sits in Tata Motors Passenger Vehicles Ltd (TMPV), while the commercial vehicle business is listed as Tata Motors Ltd. Shareholders received one share of the new entity for each share they held (Business Standard, 2025). Since this study looks at FY2022 to FY2024, the period before the split, the figures refer to the company as a single group.

1.3 Growth and Global Presence of Tata Motors

Growth, Milestones and EV Development

Tata Motors grew in steps. After the Indica in 1998, the next big move was outside India. The company bought Daewoo's commercial vehicle business in South Korea and the Spanish bus maker Hispano Carrocera, and in 2008 it acquired Jaguar Land Rover from Ford for about US$2.3 billion (Trivedi, 2014). That deal made Tata Motors a global company, but it also added debt and exposed the company to the ups and downs of the luxury car market.

In recent years the company has put a lot of effort into electric vehicles. In FY2024 it sold 73,800 electric vehicles, 48% more than the year before, and its cumulative electric vehicle production crossed 150,000 (Mobility Outlook, 2024). FY2024 was also the year in which consolidated revenue reached a record, and the company said it was on track to become net automotive debt free on a consolidated basis in FY2025.

International Operations, Subsidiaries and Partnerships

Jaguar Land Rover is the largest part of Tata Motors' international business. It designs and builds vehicles mainly in the United Kingdom and sells them in markets such as Europe, North America and China. Apart from JLR, the company has had joint ventures and subsidiaries abroad, for example a joint venture in Thailand with Thonburi Automobile (Divya & Abirami, 2016) and the Korean and Spanish businesses mentioned above. Reports in 2025 also pointed to plans for the commercial vehicle business to acquire Iveco of Italy (Moneycontrol, 2025).

Operating in many countries brings its own problems. Exchange rates move between the pound, the dollar and the rupee, import duties and emission rules differ from one market to another, and supply chains can be disrupted, as the chip shortage showed. The dataset used in this study covers sales in seven countries (India, the USA, the UK, Germany, France, Australia and Japan), which reflects this international spread.

1.4 Financial Performance of Tata Motors

Revenue, Profitability and Cost Structure

FY2024 was the best year in the company's history. Consolidated revenue was ₹437,928 crore, which was 26.6% higher than the previous year, and profit before tax rose from ₹1,803 crore in FY2023 to ₹28,932 crore (Tata Motors, 2024). Jaguar Land Rover alone earned £29.0 billion, up 27% on the year before. The commercial vehicle business also had its highest revenue and profit in FY2024.

The main costs for a vehicle maker are raw materials such as steel and aluminium, parts bought from suppliers, employee costs, and spending on research and marketing. When these costs rise faster than prices, margins fall. Interest on borrowings also takes a share of profit, which is why reducing debt has been such an important target for the company.

Cash Flow and Key Factors Influencing Performance

Cash flow shows whether the profit on paper is actually turning into cash. The company reported its highest ever free cash flows in FY2024, and it used part of this to repay debt (Tata Motors, 2024). At the same time, large amounts of money go into capital expenditure for new models, electric vehicle platforms and plant upgrades.

Several factors explain the swings in performance: how JLR's sales and product mix are doing, the availability of chips and other parts, commodity prices, exchange rates, demand in India, government policy on electric vehicles, and interest rates. The demerger in 2025 is likely to affect how the two businesses report their results in future years.

1.5 Role of Business Analytics in Financial Analysis

Excel and SQL for Data Analysis

Excel is usually the first tool used for financial data. It is good for cleaning records, removing columns that are not needed, calculating ratios such as profit margin, and building pivot tables and simple charts. Its limit is size and repeatability, since very large or very messy datasets become slow and hard to check.

SQL is used to ask questions of data kept in tables. With a few basic commands such as SELECT, WHERE, GROUP BY and ORDER BY, it is possible to find total revenue by year, rank countries by profit or filter out low margin records. The same query can be run again on new data and gives the same result, which makes the analysis easy to repeat and to check.

Tableau for Visualisation and Reporting

Tableau turns tables into an interactive dashboard. A reader can use slicers to choose a year or a country, look at key figures on cards, and see trends in charts without opening the raw data. Studies have found that tableau helps in faster reporting and better control over financial information (Khan & Jaf, 2026). In this project it is used at the end to present the main findings in one place.

1.6 Problem Statement

Tata Motors has published strong results for FY2024, but a headline figure does not explain where the profit came from, which markets contributed most, or whether costs were controlled. Many existing studies on the company rely only on ratios calculated from annual reports. There is less work that takes detailed sales and financial records and works through them with a set of analytics tools. This study tries to fill that gap by asking how Excel, SQL and Power BI can be used together to understand the financial performance of Tata Motors between FY2022 and FY2024.

1.7 Objectives of the Study

Primary Objective

To analyse the financial performance and trends of Tata Motors over the selected three-year period, FY2022–FY2024, using business analytics tools.

Secondary Objectives

1. To examine the trends in revenue, units sold and profit during FY2022–FY2024.
2. To evaluate the company's profitability using financial measures such as net profit margin, gross margin, EBITDA and EBIT, wherever the required data is available.
3. To compare revenue, profit and profitability across the seven countries represented in the dataset.
4. To calculate and analyse relevant financial ratios and year-to-year growth rates to understand financial performance trends.
5. To use Excel, SQL and tableau to organise, analyse and visualise financial data.
6. To identify significant financial trends and provide recommendations based on the findings.

1.8 Research Questions

1. How did revenue, units sold and profit change during FY2022–FY2024?
2. How did the company's profitability measure, particularly net profit margin, change over the selected period?
3. Which countries represented in the dataset recorded the highest revenue and profit, and which recorded the lowest profitability?
4. What year-on-year growth trends can be identified in revenue, units sold and profit?
5. How effectively can Excel, SQL and Tableau support the analysis and visualisation of financial performance data?

1.9 Scope of the Study

The study covers three financial years, FY2022, FY2023 and FY2024, with the financial year running from April to March. It uses a cleaned secondary dataset of 1,050 records, with 25 variables, covering 50 vehicle models and 7 countries. [State the source of the dataset here, for

example the website or file it came from.] The analysis is limited to revenue, units sold, profit, margins, expenses, tax, cash flow, capital expenditure and dividends. The tools used are Excel for cleaning and ratio calculation, basic SQL for queries, and Power BI for the dashboard.

1.10 Limitations of the Study

1. The study covers only three years, so long-term trends cannot be judged from it.
2. It relies on secondary data. The dataset may not match the audited consolidated figures in the company's annual reports, so the results describe the dataset and should not be read as an exact account of Tata Motors' reported accounts.
3. The dataset has no balance sheet items, so liquidity and debt ratios could not be calculated.
4. In the dataset, gross margin, EBITDA and EBIT are very close to net revenue, so net profit margin is used as the main measure of profitability.
5. The demerger of 2025 is outside the study period and is not analyzed.
6. Only basic SQL is used, and no forecasting or comparison with other companies is attempted.

1.11 Significance of the Study

For students, the study shows a complete analytics process on real business data, from cleaning to reporting. For managers and investors, it shows how a few free or common tools can turn a large table into clear answers about revenue, profit and costs. For future researchers, it gives a simple method that can be repeated for other companies or longer periods.

1.12 Structure of the Report

The report has five chapters. Chapter 1 gives the background, objectives and scope. Chapter 2 reviews earlier studies and points out the research gap. Chapter 3 explains the data and methodology. Chapter 4 presents the analysis and interpretation using Excel, SQL, and tableau. Chapter 5 gives the findings, suggestions and conclusion.

1.13 The Rationale Behind This Project

I chose this project to learn the main steps of a data analytics process: collecting data, cleaning it, organising it, analysing it and presenting the results. Reading about these steps is not the

same as doing them, and a real business dataset shows how many small decisions are involved, such as which columns to keep and how to deal with figures that do not agree with each other.

Tata Motors was chosen because it is a large company that has been in business for many decades and has a visible, well-known financial story. A company with this history gives plenty to analyse, and the findings are easy to relate to real business decisions. The project also helps me see how the same tools could be used by a company to track sales, control costs and take decisions in everyday work.

CHAPTER 2; REVIEW OF LITERATURE

2.1 Business analytics, data-driven decision-making.

Paper 1: Oesterreich, Anton and Teuteberg (2022)

Title: What Translates Big Data into Business Value? A Meta-Analysis of the Impacts of Business Analytics on Firm Performance

Summary: The researchers conducted a meta-analysis of 125 firm-level studies from 26 countries to examine how business analytics creates value for organizations. They found that human resources, management capabilities, and organizational culture had a stronger influence on business value than technical factors alone. The study suggests that simply adopting analytical tools is not sufficient; organizations also need people and processes that can use the resulting insights effectively. This paper supports the use of business analytics as a decision-support approach in the present study.

Paper 2: Chatterjee et al. (2023)

Title: Assessing the Impact of Big Data Analytics on Decision-Making Processes, Forecasting, and Performance of a Firm

Summary: The authors examined how big data analytics affects decision-making, forecasting, and firm performance. Their study used survey data from 366 respondents in Indian firms and applied partial least squares structural equation modelling (PLS-SEM). The findings indicated that big data analytics can support smarter decisions and more accurate forecasting. This paper is relevant to the present project because it explains how analyzing historical financial data can support the identification of trends and the interpretation of business performance.

Paper 3: Cao and Duan (2017)

Title: How Do Top- and Bottom-Performing Companies Differ in Using Business Analytics?

Summary: The researchers examined 117 manufacturing companies in the United Kingdom to investigate differences in business analytics practices between higher- and lower-performing firms. They used multivariate analysis of variance to study analytics usage, data-driven working environments, and the alignment of analytics with business processes. The study found significant differences between the two groups in their use of analytics and data-driven practices. It highlights the importance of embedding analytics into business processes rather than treating it as a separate technical activity. This provides a foundation for using financial

dashboards to support systematic performance monitoring.

Paper 4: Hussinki (2022)

Title: Business Analytics and Firm Performance: A Literature Review

Summary: Henri Hussinki reviewed research on the relationship between business analytics and organizational performance. The paper discusses how the increasing availability of business data has encouraged organizations to invest in analytics, while noting that such investments do not automatically produce measurable benefits. The review highlights the need to understand the capabilities and organizational conditions that enable analytics to create value. This paper supports the present study's approach of combining analytical tools with meaningful financial indicators and interpretation.

Paper 5: Ain, DeLone and Vaia (2025)

Title: Measuring the Success of Business Intelligence and Analytics Systems: A Literature Review

Summary: The authors reviewed 173 articles published between 2000 and 2024 to examine how the success of business intelligence and analytics systems has been measured. Their review considered the different criteria used to evaluate these systems and developed a consolidated set of success measures. The paper is relevant because it shows that the value of an analytics system should be evaluated in terms of its usefulness and outcomes, not simply whether a dashboard or software platform has been implemented. In the present project, this supports evaluating whether the analysis makes financial trends easier to understand.

Paper 6: Kesuma et al. (2025)

Title: Implementation of Power BI (Business Intelligence) in Processing and Visualizing Research Data at XYZ Institution

The researchers examined the implementation of Power BI to process and visualize institutional research data. The study focused on handling historical data and presenting it through business intelligence techniques to improve monitoring, transparency, and data-driven decision-making. It provides an applied example of using Power BI to convert structured data into visual information. This paper is relevant to the present project because Power BI will be used to present financial indicators, yearly comparisons, and trends through interactive dashboards.

Connection to Excel and SQL: These six studies provide a research foundation for business

analytics, data-driven decisions, and visualization. Excel and SQL will serve as practical tools in your project for data cleaning, calculations, querying, and aggregation. The papers should not be presented as evidence that every individual tool independently improves financial performance.

2.2 Financial performance and trends of Tata Motors and the automobile industry

Paper 7: Mahajan and Sarkar (2007)

Title: How Does Financial Performance of MNCs in the Automobile Sector Compare with Indian Companies? An Analysis Using Financial Ratios

Summary: The authors compared three Indian automobile companies—Tata Motors, Maruti Suzuki, and Mahindra & Mahindra—with Honda and Hyundai. They used ten financial ratios covering profitability, liquidity, and solvency. The study demonstrates how financial ratios can be used to compare the financial position and performance of automobile companies. It provides a foundation for selecting appropriate financial indicators for the present study.

Paper 8: Venkatachalam and Kavitha (2020)

Title: Financial Performance Analysis of Indian Commercial Vehicle Industry in India: A Study with Special Reference to Mahindra & Mahindra Limited and Tata Motors Limited

Summary: The researchers compared Tata Motors and Mahindra & Mahindra over the period 2008–09 to 2018–19. They used financial ratios, mean, standard deviation, coefficient of variation, compound annual growth rate, and multiple linear regression. The study reported that Mahindra & Mahindra performed better on several measures, including net profit, return on net worth, and selected efficiency indicators. This paper demonstrates how financial ratios and statistical methods can be used to study the financial performance of automobile manufacturers over time.

Paper 9: Mahamuni and Jumle (2021)

Title: Profitability Comparison for Automobile Companies in India Using DuPont Analysis

Summary: The authors compared Tata Motors and Mahindra & Mahindra using DuPont analysis. This method examines return on equity through net profit margin, asset turnover, and the equity multiplier. The researchers also used return on investment measures and a t-test to compare financial performance over a five-year period. The paper is useful for understanding

the factors that influence profitability and for identifying suitable financial measures for a company-level performance study.;

Paper 10: Kanagaraj and Priyadharshini (2021)

Title: A Study on Financial Performance of Tata Motors

Summary: The authors analyzed Tata Motors' financial performance over six years, from 2015 to 2020, using secondary data obtained from the company's annual reports, balance sheets, and profit-and-loss statements. The study focused on financial analysis and profitability ratios to understand the company's past performance and financial position. It emphasizes the importance of examining historical financial statements to identify strengths and weaknesses. This paper is closely related to the present study because both examine Tata Motors' financial performance over multiple years.

Paper 11: Devaki (2022)

Title: Ratio Analysis: A Study on Financial Performance of Tata Motors

Summary: Devaki examined Tata Motors' financial performance from 2017 to 2021 using secondary financial data. The study focused on three ratios: net profit margin, fixed asset turnover, and current ratio. These measures were used to assess profitability, asset utilization, and short-term financial position. The paper is relevant to the present project because it demonstrates how selected financial ratios can be used to evaluate trends in Tata Motors' performance over a five-year period.

Paper 12: Tyagi and Mahajan (2022)

Title: What Determines Profitability in the Indian Automobile Industry?

Summary: The authors investigated factors associated with profitability in India's automobile and ancillary industries using financial data covering 1999–2019. They applied a two-step system generalized method of moments regression model. The study examined the roles of factors such as past profitability, operational efficiency, marketing intensity, and market conditions. Its findings indicate that profitability is influenced by multiple business and operating factors rather than by revenue alone. This research supports examining profitability alongside costs, operating expenses, and other financial indicators in the present study.

Shilpi Tyagi, Varun Mahajan, 2022

Paper 13: Elangovan and Ramasamy (2022)

Title: An Analytical Study on Ratios Influencing Profitability of Selected Indian Automobile Players

Summary: The authors examined financial ratios associated with profitability among selected Indian automobile companies. The study focused on the use of financial ratios to understand the relationship between different aspects of a company's financial position and its profitability. It is relevant to the present project because it provides an industry-level basis for selecting financial indicators and examining their relationships. The findings should be interpreted in relation to the companies and period included in the original study.

Paper 14: Kaur, Gupta and Jain (2023)

Title: Financial Ratio Analysis of an Automobile Industry – Tata Motors

Summary: The authors examined Tata Motors' financial performance over the 2019–2022 period using secondary data from annual reports and other financial sources. The study applied financial ratio analysis and descriptive methods to assess the company's financial position. It discussed the company's financial difficulties during the period and the possibility of recovery. This paper is directly relevant because it uses historical financial information to assess Tata Motors' financial performance and provides a reference for interpreting changes across years.

Paper 15: Kumar, Yadav, Kaur and Singh (2024)

Title: Mapping the Intersection of Liquidity and Profitability in Indian Automobile Industry: A Panel Data Approach

Summary: The researchers examined the relationship between liquidity and profitability among ten listed Indian automobile manufacturers using data covering April 2010 to March 2022. They applied panel data analysis to investigate the association between liquidity measures and profitability indicators. The study reported that the effect of liquidity on profitability was statistically insignificant during the period examined. This finding is useful because it shows that stronger liquidity does not necessarily guarantee higher profitability. It supports analyzing different financial indicators separately before drawing conclusions about overall performance.

2.3 Overall summary of the literature review

The first six papers establish the importance of business analytics, data quality, analytical capabilities, and data-driven decision-making. They indicate that organizations can benefit from analytics when technical tools are combined with appropriate skills, processes, and

interpretation. They also demonstrate the role of business intelligence systems and dashboards in presenting complex information in an understandable form.

The remaining nine papers focus on financial performance in Tata Motors and the Indian automobile industry. They examine financial ratios, profitability, liquidity, asset utilization, and changes in financial performance over time. These studies provide a foundation for selecting suitable financial indicators and applying trend analysis to historical data.

Taken together, the literature supports an approach that combines financial analysis with business analytics tools. Excel and SQL can be used to prepare, calculate, and summarize financial data, while Power BI can present the results through interactive dashboards.

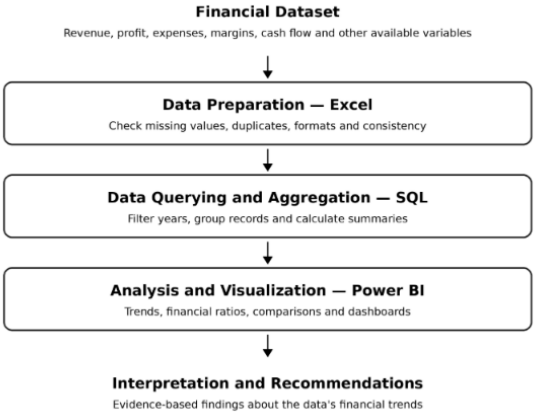
2.4 Research gap

Previous studies have examined Tata Motors' financial performance through ratio analysis, profitability measures, and comparisons with other automobile companies. Other studies have investigated how business analytics and data visualization support organizational decision-making.

However, the proposed project focuses on combining these approaches by examining financial trends over a selected 3-year period and presenting the results using business analytics tools. It will analyze relevant financial indicators, identify patterns and variations across years, and develop an interactive dashboard to communicate the findings.

The project will therefore provide a practical application of business analytics to financial performance analysis. Its contribution will depend on the quality, authenticity, and suitability of the selected dataset.

2.5 CONCEPTUAL FRAMEWORK (DIAGRAM)



CHAPTER 3: RESEARCH METHODOLOGY

3.1 RESEARCH DESIGN

This study uses a descriptive and analytical research design. It is descriptive because it presents what the data shows about revenue, units sold and profit in each year and each country. It is analytical because it goes further and calculates ratios and growth rates, compares countries and ranks records to explain the patterns behind the totals. The approach is quantitative, since all the variables are numerical, and it is based on secondary data. The study is also a single case study, with Tata Motors as the company under study.

3.2 SOURCES AND TYPES OF DATA

The study uses secondary data, meaning data that was already collected and recorded by someone else and not gathered by the researcher through surveys or interviews. The dataset contains sales and financial records of Tata Motors. [State the source here, for example the website, platform or file it was taken from, and the date it was accessed.]

The data is quantitative and structured in rows and columns. Each row is one record for a vehicle model, in one country, in one year. After cleaning, the dataset has 25 variables, which can be grouped as follows:

- **Identifiers:** Finance ID, Model ID, Year and Country.
- **Sales:** Units Sold and Total Revenue.
- **Profitability:** Profit, Profit Margin (%), Gross Margin, EBITDA, EBIT and Net Profit.
- **Costs and other items:** Marketing Expense, R&D Expense, Operating Expense, Interest Expense, Tax Rate, Tax Paid, Depreciation, Amortization and Dividend.
- **Cash and investment:** Cash Flow, Capital Expenditure and Shares Outstanding.

The columns labelled (USD) are in US dollars, and this study uses them for all comparisons.

3.3 DATA COLLECTION PROCEDURE

The data was obtained as a single CSV file containing 3,500 records for the years 2015 to 2024, with 35 columns. It covers 50 vehicle models and seven countries: India, the USA, the UK, Germany, France, Australia and Japan, with 350 records for each year. The file was first opened in Microsoft Excel, where it was cleaned and analysed. The cleaned data was then imported

into MySQL Workbench for SQL queries, and the Excel workbook was connected to Tableau to build the dashboard. Using the same cleaned dataset in all three tools keeps the results consistent.

3.4 STUDY PERIOD AND SAMPLE SELECTION

The study covers FY2022, FY2023 and FY2024. This period was chosen because it is the recent period after the pandemic, and it matches the years in which Tata Motors reported its recovery and record results (Tata Motors, 2024). The dataset has a Year column only, so the year given in the data is treated as the financial year label.

Records for 2015 to 2021 were removed. No sampling was done within the chosen period. All 350 records for each of the three years were kept, which gives 1,050 records in total, covering all 50 models and all seven countries. The three-year period is short, and this is stated as a limitation in Chapter 1.

3.5 DATA CLEANING AND PREPARATION

The data was prepared in Excel in the following steps.

1. **Filtering:** only records for 2022, 2023 and 2024 were kept (3,500 reduced to 1,050).
2. **Removing unwanted columns:** ten columns were removed because they were not needed for the objectives. These were minimum price, maximum price, revenue per unit, import duty, local manufacturing cost, warranty cost, dealer commission, currency, exchange rate and net income. The currency column was removed because it did not match the country (for example, some Indian records were marked with a different currency). Net income was removed because it nearly duplicated Net Profit, which was kept.
3. **Checking quality:** the data was checked for missing values and duplicate rows, and none were found.
4. **Formatting:** column headers were renamed in plain words, money values were rounded to two decimals, country names were standardised (for example, UK and USA), and the rows were sorted by year, country and model.
5. **Table format:** the data was converted into an Excel table with light colours so that it is easy to read and filter.

An additional step was needed in SQL. When the data was imported into MySQL, the number columns were stored as text with commas, for example 285,209,044. MySQL reads only the digits before the first comma, which gave wrong totals at first. To correct this, a view named tata_clean was created that removes the commas and converts the values to numbers before any calculation. The corrected results were then checked against the Excel totals and matched.

3.6 DATA ANALYSIS TECHNIQUES

Only simple and widely used techniques were applied.

- ✓ **Descriptive statistics:** count, mean, median, minimum, maximum and standard deviation of units sold, net revenue, net profit and profit margin.
- ✓ **Aggregation:** totals of units, revenue and profit by year and by country.
- ✓ **Ratio analysis:**
    - Net Profit Margin (%) = Net Profit ÷ Net Revenue × 100
    - Gross Margin (%) = Gross Margin ÷ Net Revenue × 100
    - EBITDA Margin (%) = EBITDA ÷ Net Revenue × 100
    - EBIT Margin (%) = EBIT ÷ Net Revenue × 100
- ✓ **Growth analysis:** Year-on-year growth (%) = (Current year − Previous year) ÷ Previous year × 100, calculated for net revenue, units sold and net profit.
- ✓ **Comparison and ranking:** countries were compared on net profit and margin, and models and country-year combinations were ranked by profit.
- ✓ **Visual analysis:** bar charts, a line chart, a map and a heatmap were used to show trends and differences.

In this dataset, gross margin, EBITDA and EBIT are almost equal to net revenue, so their margins come out near 100%. For this reason, Net Profit Margin is used as the main measure of profitability. The dataset also contains a record-level Profit Margin (%) column, which averages about 17%. This is different from the calculated Net Profit Margin, which is about 4.45%, so the two are named separately throughout the report.

3.7 TOOLS USED: EXCEL, SQL AND TABLEAU

Each tool was given a different task, so that the work was not repeated.

Tool	Purpose in this study	Main output
Microsoft Excel	Cleaning, formatting, formulas (SUMIFS), ratios, growth rates, descriptive statistics, conditional formatting	Clean_Data, Year_Summary, Country_Summary and Descriptive_Stats sheets
SQL (MySQL Workbench)	Querying the data to answer specific questions	Four queries on top models, loss-making records, countries above the overall margin and top country-year combinations
Tableau	Presenting the findings visually and interactively	A dashboard with key figures, charts, a map and a heatmap

Excel was used for the main calculations, using the SUMIFS formula to total values by year and country.

SQL was used with the commands SELECT, WHERE, GROUP BY, HAVING, ORDER BY and LIMIT, and the functions SUM, COUNT and ROUND. The queries identified the five most profitable models, the number of loss-making records in each year, the countries performing above the overall margin, and the three strongest country-year combinations.

Tableau was used to build dashboard.

3.8 ETHICAL CONSIDERATIONS

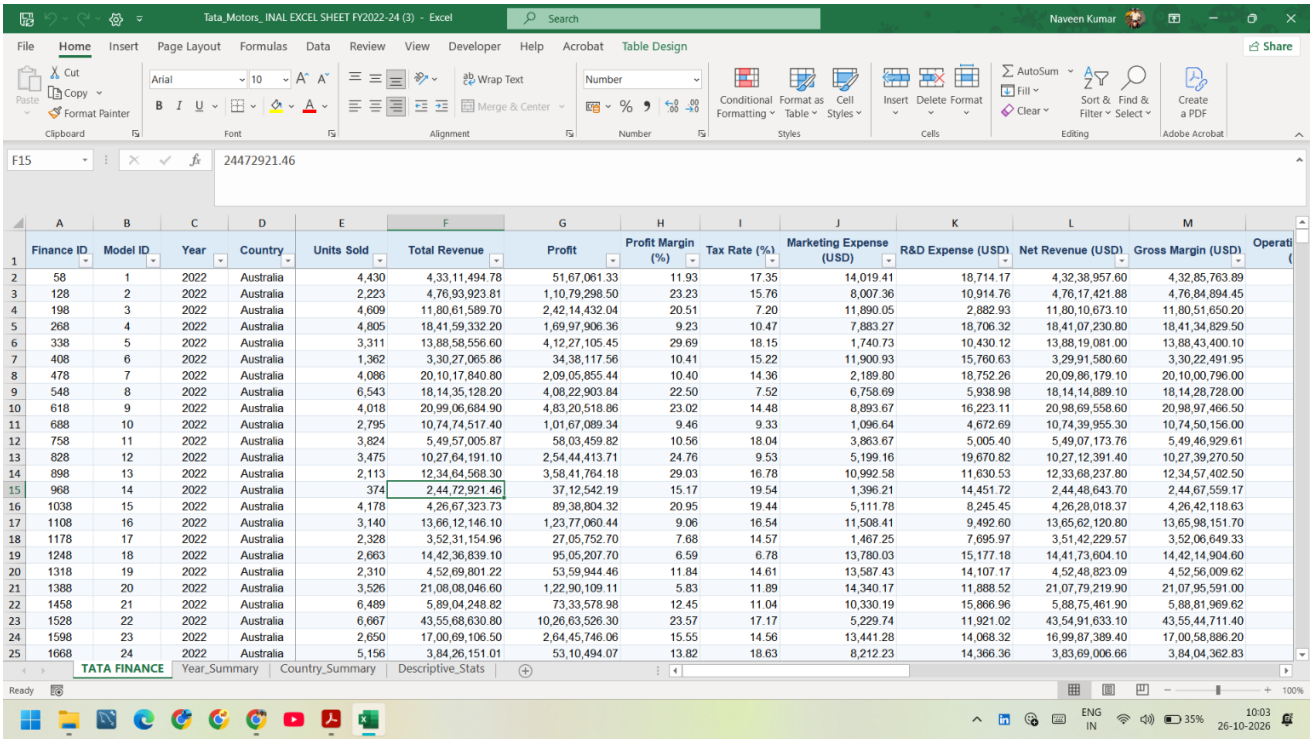
The study uses secondary business data, so no people were surveyed or interviewed. The dataset contains financial and sales records only, with no personal or customer information. The data source is acknowledged in the report, and all other references are cited. The cleaning steps and calculations are described openly so that the work can be repeated and checked. Because the dataset may not match the audited figures in the annual reports of Tata Motors, the results are presented as findings about the dataset and not as an official account of the company's reported performance. The software was used as a standard installed or free edition, and the analysis and interpretation are the researcher's own work.

CHAPTER 4: DATA ANALYSIS AND INTERPRETATION

4.1 INTRODUCTION TO THE DATASET

This chapter explains the analysis of the Tata Motors dataset using Excel, SQL and Tableau. The cleaned dataset contains 1,050 records and 25 variables for the financial years 2022, 2023 and 2024. Each record shows the financial result of one vehicle model in one country in one year. The data covers 50 models and seven countries: India, the USA, the UK, Germany, France, Australia and Japan. Each year has 350 records.

Figure 4.1 shows the cleaned dataset in Excel.



Finance ID	Model ID	Year	Country	Units Sold	Total Revenue	Profit	Profit Margin (%)	Tax Rate (%)	Marketing Expense (USD)	R&D Expense (USD)	Net Revenue (USD)	Gross Margin (USD)
58	1	2022	Australia	4,430	4,33,11,494.78	51,67,061.33	11.93	17.35	14,019.41	18,714.17	4,32,38,957.60	4,32,85,763.89
128	2	2022	Australia	2,223	4,76,93,923.81	1,10,79,298.50	23.23	15.76	8,007.36	10,914.76	4,76,17,421.88	4,76,84,894.45
198	3	2022	Australia	4,609	11,80,61,589.70	2,42,14,432.04	20.51	7.20	11,890.05	2,882.93	11,80,10,673.10	11,80,51,650.20
268	4	2022	Australia	4,805	18,41,59,332.20	1,69,97,906.36	9.23	10.47	7,883.27	18,706.32	18,41,07,230.80	18,41,34,829.50
338	5	2022	Australia	3,311	13,88,58,556.60	4,12,27,105.45	29.69	18.15	1,740.73	10,430.12	13,88,19,081.00	13,88,43,400.10
408	6	2022	Australia	1,362	3,30,27,065.86	34,38,117.56	10.41	15.22	11,900.93	15,760.63	3,29,91,580.60	3,30,22,491.95
478	7	2022	Australia	4,086	20,10,17,840.80	2,09,05,855.44	10.40	14.36	2,189.80	18,752.26	20,09,86,179.10	20,10,00,796.00
548	8	2022	Australia	6,543	18,14,35,128.20	4,08,22,903.84	22.50	7.52	6,758.69	5,938.98	18,14,14,889.10	18,14,28,728.00
618	9	2022	Australia	4,018	20,99,06,684.90	4,83,20,518.86	23.02	14.48	8,893.67	16,223.11	20,98,69,558.60	20,98,97,466.50
688	10	2022	Australia	2,795	10,74,74,517.40	1,01,67,089.34	9.46	9.33	1,096.64	4,672.69	10,74,39,955.30	10,74,50,156.00
758	11	2022	Australia	3,824	5,49,57,005.87	58,03,459.82	10.56	18.04	3,863.67	5,005.40	5,49,07,173.76	5,49,46,929.61
828	12	2022	Australia	3,475	10,27,64,191.10	2,54,44,413.71	24.76	9.53	5,199.16	19,670.82	10,27,12,391.40	10,27,39,270.50
898	13	2022	Australia	2,113	12,34,64,568.30	3,58,41,764.18	29.03	16.78	10,992.58	11,630.53	12,33,68,237.80	12,34,57,402.50
968	14	2022	Australia	374	2,44,72,921.46	37,12,542.19	15.17	19.54	1,396.21	14,451.72	2,44,48,643.70	2,44,67,559.17
1038	15	2022	Australia	4,178	4,26,67,323.73	89,38,804.32	20.95	19.44	5,111.78	8,245.45	4,26,28,018.37	4,26,42,118.63
1108	16	2022	Australia	3,140	13,66,12,146.10	1,23,77,060.44	9.06	16.54	11,508.41	9,492.60	13,65,62,120.80	13,65,98,151.70
1178	17	2022	Australia	2,328	3,52,31,154.96	27,05,752.70	7.68	14.57	1,467.25	7,695.97	3,51,42,229.57	3,52,06,649.33
1248	18	2022	Australia	2,663	14,42,36,839.10	95,05,207.70	6.59	6.78	13,780.03	15,177.18	14,41,73,604.10	14,42,14,904.60
1318	19	2022	Australia	2,310	4,52,69,801.22	53,59,944.46	11.84	14.61	13,587.43	14,107.17	4,52,48,823.09	4,52,56,009.62
1388	20	2022	Australia	3,526	21,08,08,046.60	1,22,90,109.11	5.83	11.89	14,340.17	11,888.52	21,07,79,219.90	21,07,95,591.00
1458	21	2022	Australia	6,489	5,89,04,248.82	73,33,578.98	12.45	11.04	10,330.19	15,866.96	5,88,75,461.90	5,88,81,969.62
1528	22	2022	Australia	6,667	43,55,68,630.80	10,26,63,526.30	23.57	17.17	5,229.74	11,921.02	43,54,91,633.10	43,55,44,711.40
1598	23	2022	Australia	2,650	17,00,69,106.50	2,64,45,746.06	15.55	14.56	13,441.28	14,068.32	16,99,87,389.40	17,00,58,886.20
1668	24	2022	Australia	5,156	3,84,26,151.01	53,10,494.07	13.82	18.63	8,212.23	14,366.36	3,83,69,006.66	3,84,04,362.83

The main variables used in the analysis are Year, Country, Model ID, Units Sold, Net Revenue (USD), Gross Margin (USD), EBITDA (USD), EBIT (USD) and Net Profit (USD). All amounts in this chapter are in US dollars. No missing values or duplicate records were found during cleaning.

Two points about the data should be kept in mind while reading the results:

1. Gross margin, EBITDA and EBIT are almost equal to net revenue in this dataset, so their margins come out near 100%. For this reason, net profit margin is used as the main measure of profitability.
2. The dataset also has a record-level Profit Margin (%) column, which averages about 17%. This is different from the net profit margin calculated in this study (Net Profit ÷ Net Revenue, about 4.45%), so the two are named separately.

4.2 DESCRIPTIVE ANALYSIS OF FINANCIAL DATA

Descriptive statistics help explain the usual values in the dataset and how much the values differ. They were calculated in Excel for all 1,050 records.

Table 1: Descriptive statistics (per record)

Variable	Count	Mean	Median	Minimum	Maximum	Std. Deviation
Units Sold	1,050	3,539	3,557	104	6,956	2,024
Net Revenue (USD)	1,050	132,948,721	107,374,575	976,624	472,612,971	106,648,091
Net Profit (USD)	1,050	5,922,754	2,753,344	−54,935,399	87,270,176	14,412,934
Profit Margin (%)	1,050	17.24	17.02	5.01	29.94	7.25

Interpretation

- Units sold: the mean (3,539) and median (3,557) are close, so sales are spread fairly evenly. The range is wide, from 104 to 6,956 units per record.
- Net revenue: the mean ($132.9 million) is higher than the median ($107.4 million). This means a few large records pull the average up, and most records are below the average.
- Net profit: the standard deviation ($14.4 million) is more than twice the mean ($5.9 million), which shows very large variation between records. The minimum is negative (−$54.9 million), so some records made a loss, and the median ($2.8 million) is less than half of the mean.
- Profit margin (%): the mean (17.24) and median (17.02) are almost the same, and the values stay between 5.01 and 29.94.

Overall, most records show a profit, but results vary widely. Some records earn high profits, while others report losses.

4.3 YEAR-WISE REVENUE AND PROFIT TRENDS

Totals for each year were calculated in Excel using the SUMIFS formula. They answer Research Question 1.

Table 2: Year-wise summary (USD)

Metric	2022	2023	2024	Total
Units Sold	1,234,585	1,256,786	1,224,670	3,716,041
Net Revenue	47,237,449,174	47,520,171,723	44,838,536,541	139,596,157,438
Net Profit	1,944,401,414	1,988,405,577	2,286,084,831	6,218,891,821

Table 3: Year-on-year growth

Measure	2023	2024
Net Revenue growth	+0.60%	−5.64%
Units Sold growth	+1.80%	−2.56%
Net Profit growth	+2.26%	+14.97%

Interpretation

- Revenue: net revenue was about $47.24 billion in 2022 and rose slightly to $47.52 billion in 2023 (+0.60%). In 2024 it fell to $44.84 billion (−5.64%), a drop of about $2.68 billion. Compared with 2022, revenue in 2024 was about 5.1% lower.
- Units sold: units increased by 1.80% in 2023 and then fell by 2.56% in 2024. The movement of units is in the same direction as revenue in both years, which suggests that the fall in 2024 revenue is linked to lower sales volume.
- Net profit: net profit rose in both years, from $1.94 billion to $1.99 billion (+2.26%) in 2023 and to $2.29 billion (+14.97%) in 2024. Over the whole period it grew by about 17.6%.
- Net profit rose in 2024 even though revenue and units fell. This means Tata Motors earned more net profit from each dollar of revenue. The net profit margin in the next section shows this improvement.

4.4 PROFITABILITY AND FINANCIAL RATIO ANALYSIS

Profitability ratios were calculated in Excel from the yearly totals. This section answers Research Question 2.

Table 4: Profitability ratios

Ratio	2022	2023	2024	Overall
Net Profit Margin (%)	4.12%	4.18%	5.10%	4.45%
Gross Margin (%)	100.03%	100.03%	100.03%	100.03%
EBITDA Margin (%)	100.01%	100.01%	100.01%	100.01%
EBIT Margin (%)	100.00%	100.00%	100.00%	100.00%

Interpretation

- Net profit margin rose from 4.12% in 2022 to 4.18% in 2023 and 5.10% in 2024. The increase of about 0.98 percentage points over the period shows that the company kept more profit from each dollar of revenue. The biggest improvement came in 2024.
- The gross margin, EBITDA margin and EBIT margin are all close to 100% in every year. These margins are not realistic for an automobile manufacturer because vehicle production involves materials, labour and other costs. This suggests that the three values in the dataset are almost the same as net revenue, so they cannot provide a reliable picture of profitability. They were calculated to meet the study objective but are not used for conclusions. The net profit margin is the only meaningful profitability measure available in this dataset.
- Net profit margin increasing together with falling revenue in 2024 shows that the improvement in profit came from better profitability and not from higher sales.

4.5 COUNTRY-WISE ANALYSIS

This section compares the seven countries and answers Research Question 3.

Table 5: Country-wise summary (FY2022–FY2024 combined, USD)

Country	Units Sold	Net Revenue	Net Profit	Net Profit Margin
Australia	534,068	20,257,881,388	896,467,448	4.43%
France	497,721	18,316,121,650	882,545,566	4.82%
Germany	507,928	19,075,346,234	707,673,427	3.71%
India	559,649	20,910,794,375	827,592,165	3.96%
Japan	543,175	20,597,508,224	858,576,311	4.17%
UK	543,974	20,353,220,418	1,193,184,566	5.86%
USA	529,526	20,085,285,148	852,852,339	4.25%
Total	3,716,041	139,596,157,438	6,218,891,821	4.45%

Interpretation

- Revenue: India recorded the highest net revenue ($20.91 billion) and the highest units sold (559,649). France recorded the lowest net revenue ($18.32 billion). Revenue is fairly evenly spread, with six of the seven countries between $19 billion and $21 billion.
- Profit: the UK earned the highest net profit ($1.19 billion), about 19.2% of the total, although it generated only about 14.6% of the total revenue. Germany earned the lowest net profit ($0.71 billion).
- Margin: the UK had the highest net profit margin (5.86%) and Germany the lowest (3.71%), a gap of about 2.15 percentage points. Only the UK (5.86%) and France (4.82%) were above the overall margin of 4.45%.
- India has the highest revenue but one of the lowest margins (3.96%), which shows that high sales alone do not mean high profitability.

Table 4.6: Net profit by country and year (USD)

Country	2022	2023	2024	Total
Australia	384,820,815	297,527,636	214,118,997	896,467,448
France	385,078,454	188,573,661	308,893,451	882,545,566
Germany	204,650,152	82,969,694	420,053,581	707,673,427
India	234,064,562	285,931,199	307,596,403	827,592,165
Japan	124,351,893	385,973,884	348,250,535	858,576,311
UK	340,204,632	490,262,014	362,717,920	1,193,184,566
USA	271,230,905	257,167,490	324,453,944	852,852,339
Total	1,944,401,414	1,988,405,577	2,286,084,831	6,218,891,821

Interpretation

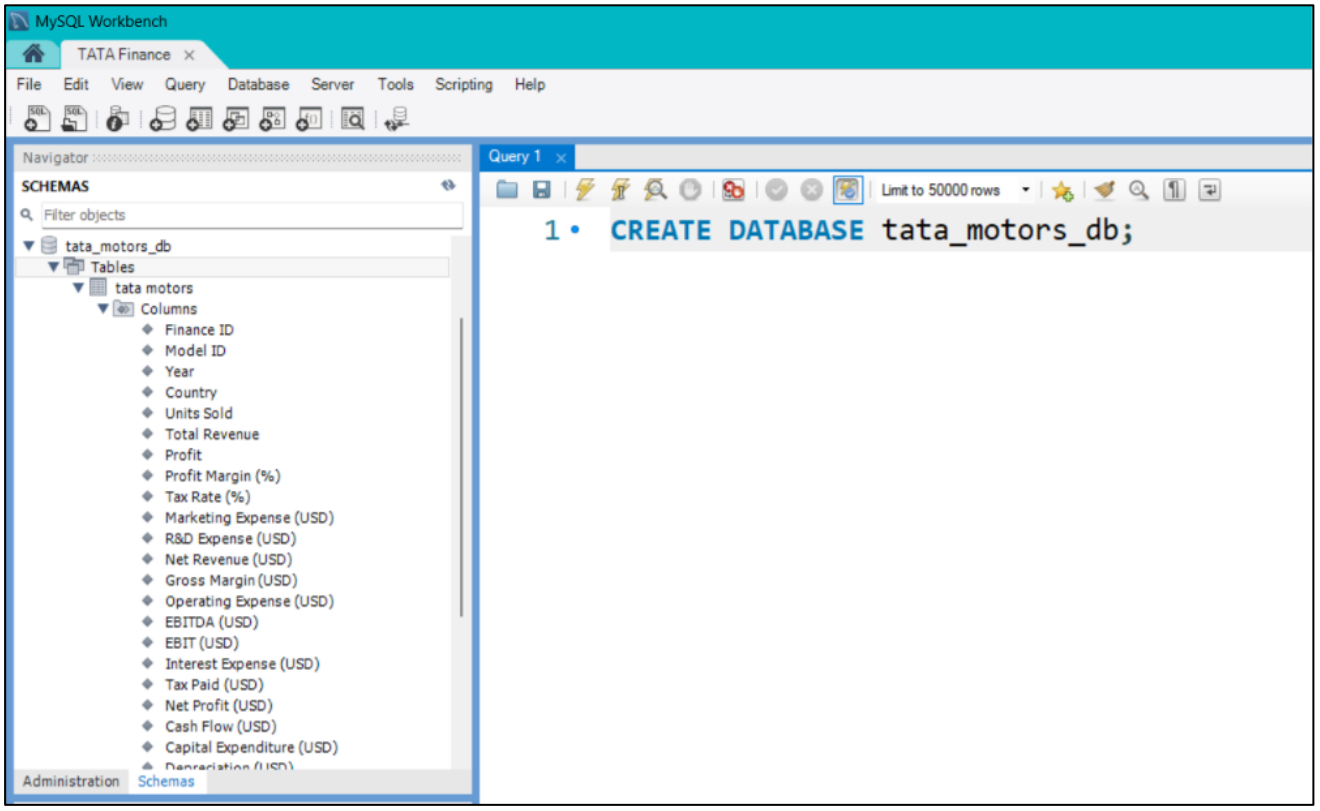
- India is the only country where net profit rose in each of the three years ($234 million, $286 million, $308 million).
- Australia is the opposite, with profit falling every year, from $385 million to $214 million.
- Germany had the lowest profit in 2023 ($83 million) and then the highest jump in 2024 ($420 million), which shows large year-to-year swings.
- The UK reached its peak in 2023 ($490 million) and fell in 2024 ($363 million). Japan rose sharply in 2023 and then eased in 2024.

The yearly totals at the bottom of the table match Table 4.2, which confirms that the figures are consistent.

4.6 SQL-BASED ANALYSIS AND AGGREGATION

SQL was used in MySQL Workbench to answer specific questions that the Excel summaries do not show. A database named tata_motors_db was created, the cleaned data was imported into a table, and a view named tata_clean was created to convert the stored text values with commas into numbers before calculation (see Section 3.5).

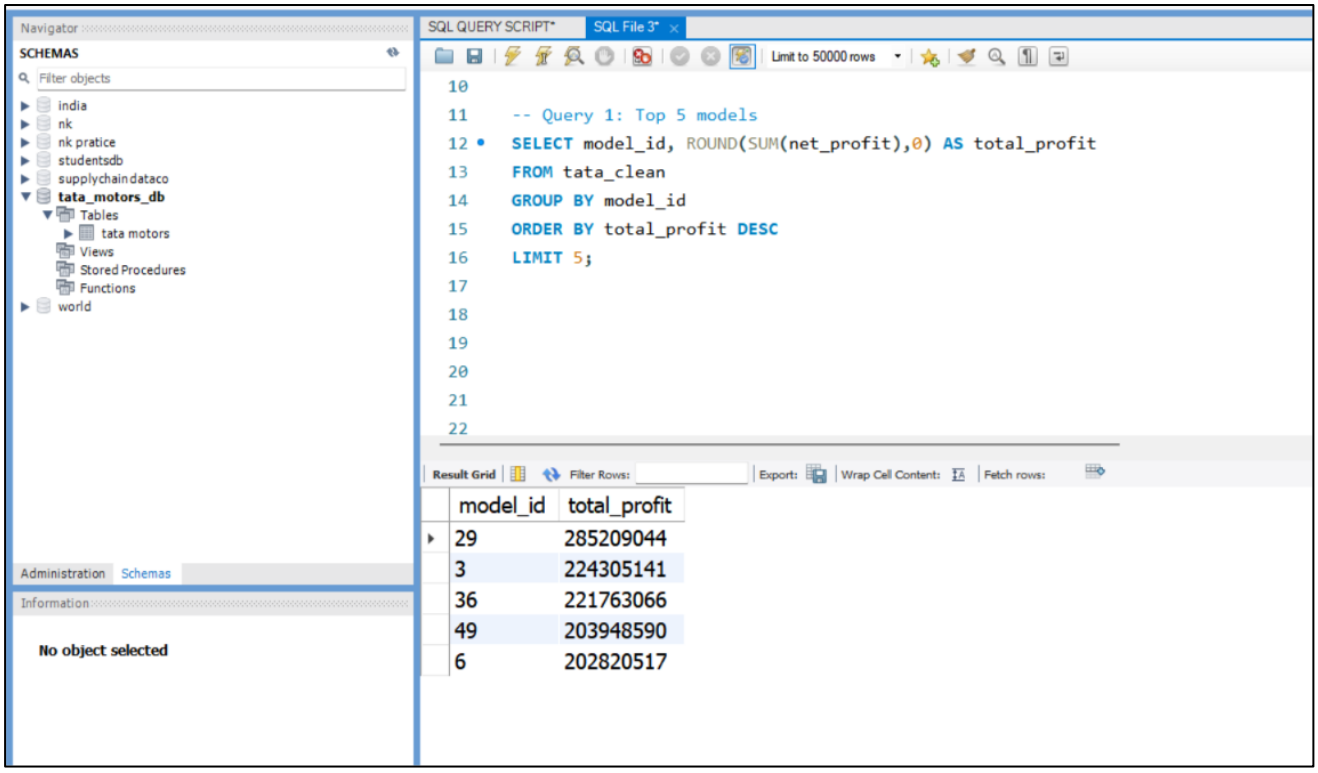
Setup: database and table creation (**Figure 1**)



Query 1: Top 5 models by net profit (**Figure 2**)

Model ID	Total Net Profit (USD)
29	285,209,044
3	224,305,141
36	221,763,066
49	203,948,590
6	202,820,517

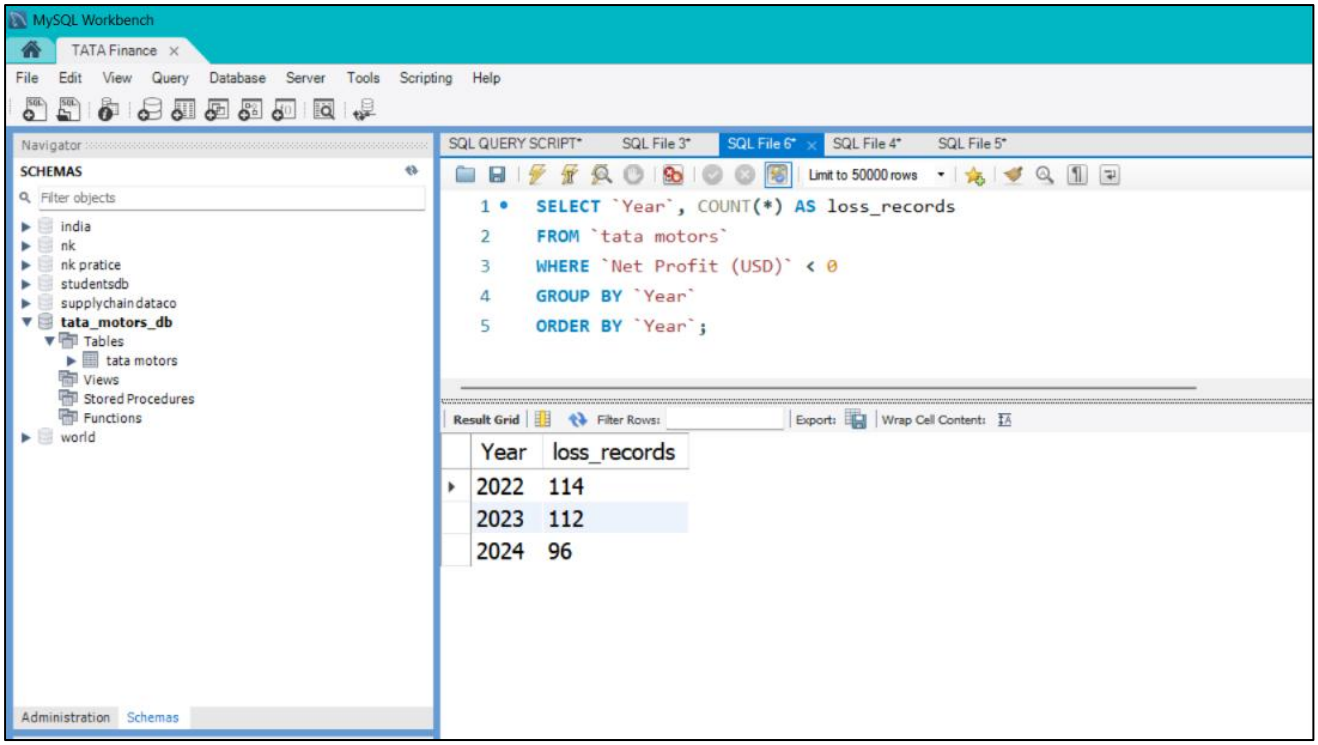
Interpretation: Model 29 earned the most profit, about $285 million (4.6% of total net profit). The five models together earned about $1.14 billion, which is 18.3% of the total, although they are only 10% of the 50 models.



Query 2: Loss-making records by year (Figure 4.5)

Year	Loss-making records
2022	114
2023	112
2024	96

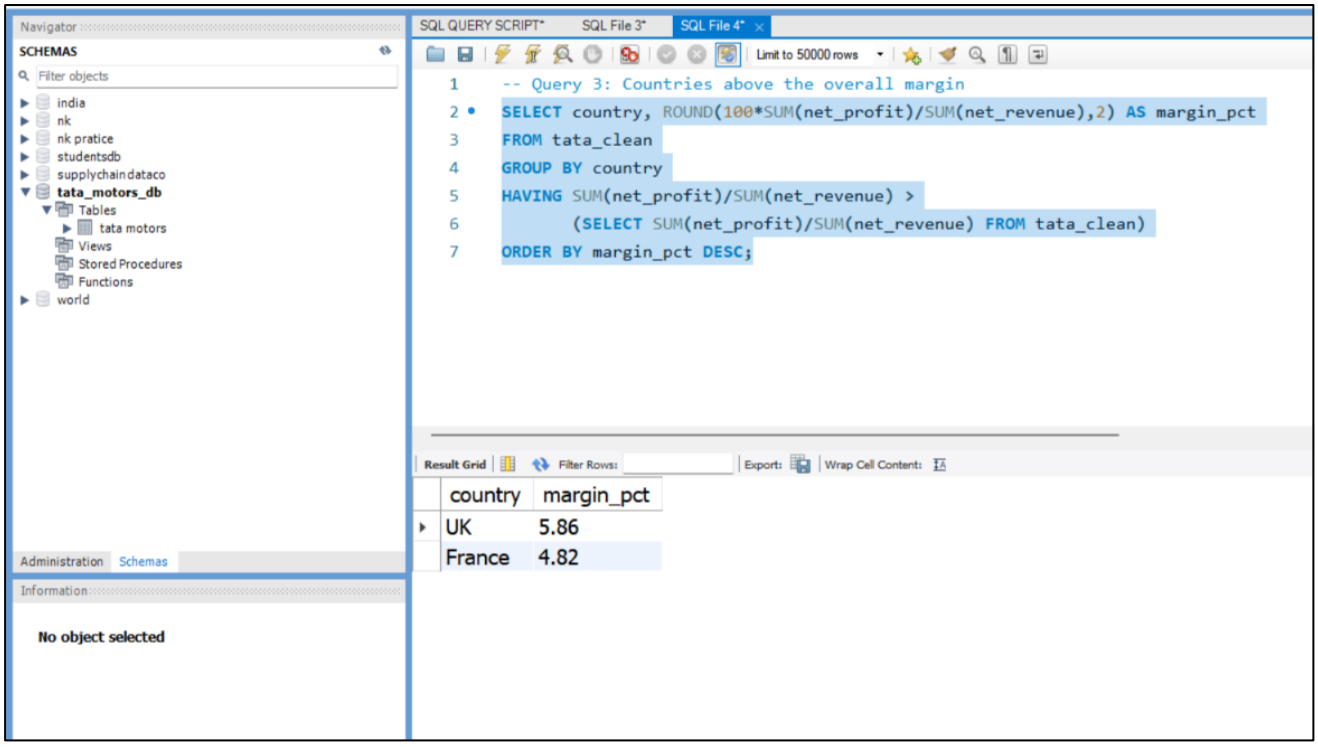
Interpretation: 322 of the 1,050 records (about 30.7%) made a loss. The number fell each year: 114 (32.6% of that year's records) in 2022, 112 (32.0%) in 2023 and 96 (27.4%) in 2024. This is consistent with the rise in net profit margin in 2024.



Query 3: Countries with a margin above the overall margin (Figure 4.6)

Country	Net Profit Margin (%)
UK	5.86
France	4.82

Interpretation: only the UK and France are above the overall margin of 4.45%. The other five countries are below it. This matches Table 4.5.

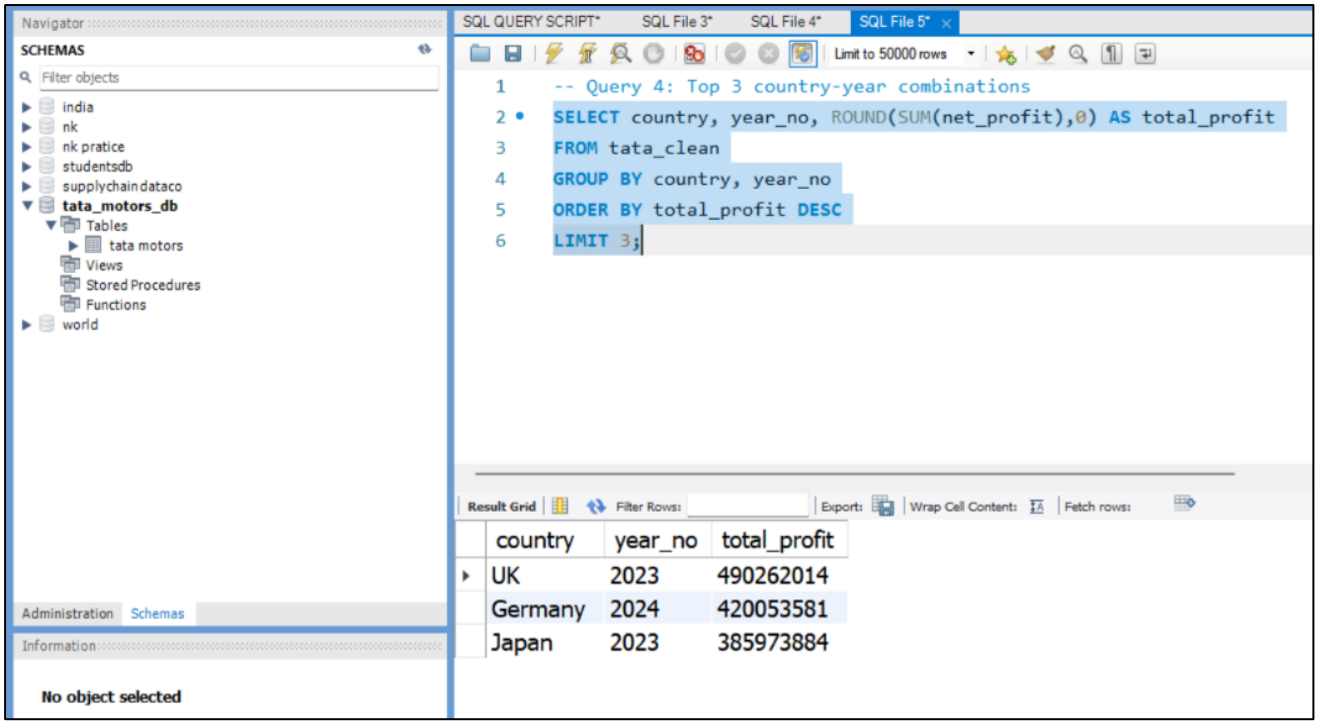


Query 4: Top 3 country-year combinations by net profit (**Figure 4.7**)

Country	**Year**	**Total Net Profit (USD)**
UK	2023	490,262,014
Germany	2024	420,053,581
Japan	2023	385,973,884

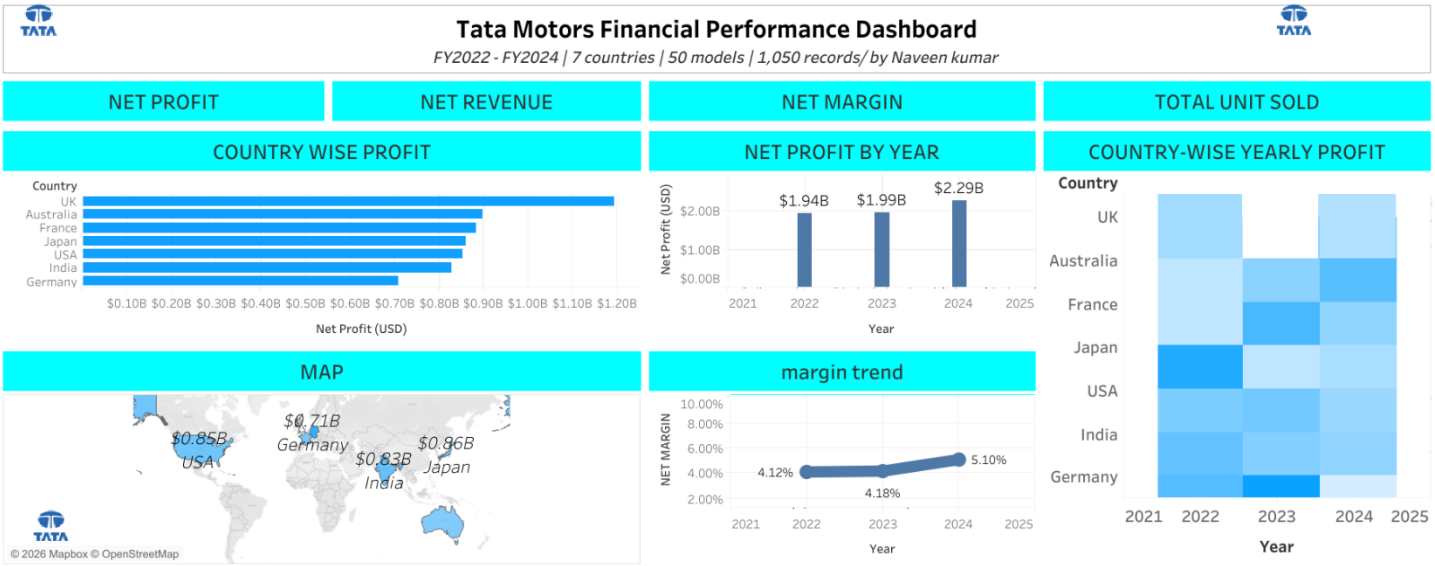
Interpretation: the strongest result was the UK in 2023 ($490 million). Germany's result in 2024 is notable, because Germany had the lowest profit of any country-year ($83 million) just one year earlier.

Cross-check: the SQL results match the Excel tables. The margins for the UK (5.86%) and France (4.82%), the country-year profits and the 322 loss-making records agree in both tools.



4.7 TABLEAU DASHBOARD AND VISUALIZATIONS

Tableau was used to present the main results in one interactive dashboard, titled “Tata Motors Financial Performance Dashboard” (**Figure 1**).



The dashboard contains the following parts:

Visual	What it shows
Key figure cards	Net Profit $6.22 billion, Net Revenue $139.60 billion, Net Margin 4.45% and Total Units Sold 3,716,041
Country-wise profit (bar chart)	Net profit by country, with the UK highest and Germany lowest
Net profit by year (column chart)	$1.94 billion, $1.99 billion and $2.29 billion for 2022 to 2024
Margin trend (line chart)	Net profit margin rising from 4.12% to 4.18% to 5.10%
Map	Net profit by country on a world map
Country-wise yearly profit (heatmap)	Net profit for each country and year, with darker colours for higher profit

Interpretation

- The key figures at the top match the Excel totals exactly, which confirms that the data was loaded correctly into Tableau.
- The bar chart and map show at a glance that the UK stands out as the most profitable country, while the other six countries are close to each other.
- The net profit chart and the margin trend show the steady improvement over the three years in a form that is easy to understand.
- The heatmap shows which country and year combinations were strongest and weakest, and supports the results in Table 4.6.
- Because the dashboard is interactive, a reader can choose a year or a country with filters and see the figures update, without opening the raw data.

4.8 FINDINGS AND INTERPRETATION

The research questions organise the findings.

RQ1: How did revenue, units sold and profit change during FY2022–FY2024? Net revenue rose by 0.60% in 2023 and fell by 5.64% in 2024. Units sold rose by 1.80% in 2023 and fell by 2.56% in 2024. Net profit increased in both years, from $1.94 billion to $1.99 billion and then $2.29 billion.

RQ2: How did the net profit margin change?

It rose from 4.12% in 2022 to 4.18% in 2023 and 5.10% in 2024. The overall margin for the period was 4.45%. Gross margin, EBITDA and EBIT were close to 100% and are not reliable measures in this dataset.

RQ3: Which countries recorded the highest revenue and profit, and which the lowest profitability?

India recorded the highest net revenue ($20.91 billion) and France the lowest ($18.32 billion). The UK recorded the highest net profit ($1.19 billion) and the highest margin (5.86%). Germany recorded the lowest profit ($0.71 billion) and the lowest margin (3.71%).

RQ4: What year-on-year growth trends can be identified?

Revenue and units sold showed the same pattern: small growth in 2023 and a decline in 2024. Net profit grew in both years, by 2.26% and 14.97%. The 2024 profit growth came with falling revenue, so it was driven by a better margin.

RQ5: How effectively can Excel, SQL and Tableau support the analysis? Each tool had a clear role. Excel was effective for cleaning the data, calculating ratios and growth rates and producing summary tables. SQL answered specific questions such as the top models, the number of loss-making records and the countries above the overall margin. Tableau presented the results in one interactive dashboard that is easy to read. The results were consistent across all three tools.

Additional findings

- About 30.7% of the records made a loss, but the number of loss-making records fell from 114 in 2022 to 96 in 2024.
- The top five models contributed 18.3% of total net profit.
- Country-level profit moved differently from year to year: India grew every year, Australia fell every year, and Germany had a sharp swing between 2023 and 2024.

These findings describe the dataset used in this study. As stated in the limitations, the dataset may not match the audited accounts of Tata Motors, so the results should not be treated as the company's reported performance.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY OF THE STUDY

This study, "An Analytical Study of the Financial Performance and Trends of Tata Motors Using Business Analytics Tools", examined a dataset of Tata Motors sales and financial records for FY2022 to FY2024. The cleaned dataset had 1,050 records and 25 variables, covering 50 vehicle models and seven countries: India, the USA, the UK, Germany, France, Australia and Japan.

The study had two aims. The first was to understand how revenue, units sold, net profit and profitability changed over the three years, and how the seven countries compared. The second was to show how Excel, SQL and Tableau can be used together to do this.

Excel was used to clean the data and to calculate yearly and country totals, ratios, growth rates and descriptive statistics. SQL, run in MySQL Workbench, was used to answer specific questions, such as which models earned the most profit and how many records made a loss. Tableau was used to present the results in one interactive dashboard. The totals matched across all three tools.

5.2 MAJOR FINDINGS

Year-wise trends

1. Net revenue rose slightly by 0.60% in 2023, from $47.24 billion to $47.52 billion, and then fell by 5.64% to $44.84 billion in 2024.
2. Units sold rose by 1.80% in 2023 and fell by 2.56% in 2024, moving in the same direction as revenue.
3. Net profit rose in both years, from $1.94 billion in 2022 to $1.99 billion in 2023 and $2.29 billion in 2024. This is a growth of about 17.6% over the period, and the largest increase (14.97%) came in 2024.

Profitability

4. Net profit margin improved from 4.12% in 2022 to 4.18% in 2023 and 5.10% in 2024. The overall margin was 4.45%.

5. Profit grew in 2024 even though revenue fell, so the improvement came from a better margin and not from higher sales.

6. Gross margin, EBITDA and EBIT were almost equal to net revenue in the dataset, so their margins were close to 100%. They did not give useful information, and net profit margin was used as the main measure.

Country-wise results

7. India recorded the highest net revenue ($20.91 billion) and France the lowest ($18.32 billion).
8. The UK earned the highest net profit ($1.19 billion, about 19.2% of the total) and the highest margin (5.86%). Germany earned the lowest net profit ($0.71 billion) and had the lowest margin (3.71%).

9. Only the UK (5.86%) and France (4.82%) were above the overall margin of 4.45%.
10. India had the highest revenue but one of the lowest margins (3.96%), so high sales did not mean high profitability.

11. Country profits moved differently over the years. India's profit rose in all three years, Australia's fell in all three years, and Germany's profit jumped from $83 million in 2023 to $420 million in 2024.

SQL results

12. Model 29 earned the most profit ($285 million), and the top five models together earned about 18.3% of total net profit.

13. 322 of the 1,050 records (30.7%) made a loss, but the number fell from 114 in 2022 to 112 in 2023 and 96 in 2024.

14. The strongest country-year result was the UK in 2023 ($490 million).

Descriptive statistics

15. Net profit varied widely between records. The standard deviation ($14.4 million) was more than twice the mean ($5.9 million), and the lowest record was a loss of $54.9 million.

Tools

16. Each tool had a clear role, and together they gave consistent results: Excel for calculation, SQL for questions and ranking, and Tableau for presentation.

These are results for the dataset used in this study. As stated in the limitations, the dataset may not match the audited accounts of Tata Motors.

5.3 RECOMMENDATIONS

The recommendations come from the findings above. Some concern the company's business, as shown in the data, and some concern how the analysis should be done.

Recommendations based on the findings

1. **Look into the fall in revenue in 2024.** Revenue fell by 5.64% and units by 2.56%. A company in this position should check which models and markets lost sales, and whether the cause was demand, supply or pricing.
2. **Keep the margin improvement.** Net profit margin rose to 5.10% in 2024. The company should find out whether this came from lower costs, better pricing or a better product mix, so that it can be repeated.
3. **Learn from the UK and study Germany.** The UK has the best margin (5.86%) and Germany the weakest (3.71%). Comparing the two markets could show which practices help profitability.
4. **Do not rely on sales volume alone.** India has the highest revenue but a below-average margin (3.96%). The company should focus on profit per sale in high-volume markets, not only on volume.
5. **Review loss-making records.** About 30.7% of the records made a loss. The number is falling, but it is still high, so these models and markets should be reviewed for price, cost or demand problems.
6. **Support the top models and watch the weak ones.** Model 29 and the other top five models bring a large share of profit. Their performance should be protected, and consistently loss-making models should be reviewed.
7. **Watch markets with unstable profit.** Germany and Australia showed large changes or steady falls in profit. These markets need closer monitoring.

Recommendations about the analysis

8. **Check the data source.** Before drawing company-wide conclusions, confirm whether the dataset is actual or modelled data, and compare it with the published annual reports.
9. **Use measures that work.** Because gross margin, EBITDA and EBIT were close to net revenue in this dataset, the net profit margin should be used for comparison unless these fields are checked and corrected.

10. **Use the same figures in every tool.** The same period, filters and definitions should be used in Excel, SQL and Tableau, as was done in this study, so that the results can be trusted.
11. **Check data types when moving data between tools.** In this study, numbers stored as text with commas gave wrong totals in SQL until they were converted. Data types should always be checked after importing.

These recommendations are based on the dataset and should not be taken as final advice to Tata Motors without checking them against the company's actual information.

5.4 CONCLUSION

This study examined the financial performance of Tata Motors for FY2022 to FY2024 using a dataset of 1,050 records and the tools Excel, SQL and Tableau.

The analysis shows that net profit increased in each year, from $1.94 billion to $2.29 billion, even though revenue and units sold fell in 2024. Net profit margin improved from 4.12% to 5.10%, which means that the company earned more profit from each dollar of revenue by the end of the period. The seven countries differed clearly in profitability. The UK was the strongest, with a margin of 5.86%, and Germany the weakest, with 3.71%, while India had the highest revenue but a below-average margin. About three in ten records made a loss, but the number of loss-making records fell over the three years.

The study also showed how the three tools can be used for different tasks in one analysis. Excel prepared the data and calculated the main measures, SQL answered specific questions and ranked results, and Tableau presented the findings clearly. Using the same cleaned data in all three tools gave consistent results, and the cross-checks between them increased confidence in the numbers.

The results describe the dataset used here. The study covers only three years, the dataset may not match the audited accounts of Tata Motors, and several margin fields were not useful. For these reasons, the findings are best seen as a worked example of financial analysis with business analytics tools, and as an indication of the trends in the data.

5.5 SCOPE FOR FUTURE RESEARCH

1. **Use verified data.** Repeat the analysis with the audited financial statements and annual reports of Tata Motors to study actual company-wide performance.
2. **Use a longer period.** Extend the study to five or ten years to see long-term trends and the effect of events such as the pandemic and the chip shortage.
3. **Compare with other companies.** Compare Tata Motors with Maruti Suzuki, Mahindra & Mahindra or Ashok Leyland using the same measures.
4. **Add balance sheet analysis.** Use balance sheet data to calculate liquidity, debt and return ratios, such as the current ratio, debt-to-equity ratio and return on equity.
5. **Study costs and cash flow.** Examine operating expenses, cash flow and capital expenditure in a separate detailed analysis.
6. **Study the demerger.** Compare the results of the passenger vehicle and commercial vehicle businesses after the 2025 demerger.
7. **Use forecasting.** Apply simple forecasting methods to estimate future revenue and profit once a longer period of data is available.
8. **Use more advanced tools.** Use Python, R or machine learning methods for deeper analysis.

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